FINANCIAL MARKET RISK & PORTFOLIO ANALYSIS

Portfolio Construction, Risk Measurement and Out-of-Sample Performance Evaluation

Research Project

Analysis period: November 2021 – November 2026
Out-of-sample evaluation: January 2025 – November 2026
Assets: AAPL, AMZN, JPM, MSFT, XOM
Benchmark: SPY

This project examines how different portfolio construction approaches affect investment returns and risk. It compares equal-weight, fixed minimum-variance and rolling minimum-variance portfolios against a market benchmark.

1. Executive Summary

This project investigates portfolio construction and risk measurement using historical daily price data for five individual stocks: Apple (AAPL), Amazon (AMZN), JPMorgan Chase (JPM), Microsoft (MSFT) and Exxon Mobil (XOM). The SPDR S&P 500 ETF (SPY) serves as the benchmark.

Three portfolio strategies were evaluated: an equal-weight portfolio, a fixed minimum-variance portfolio and a rolling minimum-variance portfolio. Their performance was compared with SPY using a monthly-rebalanced out-of-sample test.

The fixed minimum-variance strategy recorded the highest compound annual growth rate (CAGR), at 25.26%, and the highest Sharpe ratio, at 1.44. The rolling minimum-variance strategy recorded the smallest maximum drawdown, at –15.22%. These findings describe the historical test period and should not be interpreted as predictions of future returns.

Strategy	CAGR	Volatility	Sharpe	Max. Drawdown
Equal-Weight	21.82%	18.18%	1.18	–20.04%
Fixed Min-Variance	25.26%	16.65%	1.44	–16.95%
Rolling Min-Variance	21.96%	16.82%	1.26	–15.22%
SPY	14.31%	17.10%	0.87	–18.76%

2. Research Objectives

The main objectives of the project were to:

• Measure the historical returns and volatility of selected stocks.

• Examine relationships between assets using correlation analysis.

• Construct an equal-weight portfolio and a minimum-variance portfolio.

• Evaluate portfolio performance using return and risk measures.

• Test portfolio strategies on data not used for initial optimisation.

• Investigate the effects of monthly rebalancing and transaction costs.

• Examine sensitivity to different estimation-window lengths.

• Compare portfolio performance on positive and negative market days.

3. Dataset and Assets

Historical adjusted price data were collected using the Yahoo Finance data source through the Python yfinance library. The dataset covers 8 November 2021 to 6 November 2026 and contains 1,255 daily price observations for the six tickers.

Ticker	Company / Instrument	Role
AAPL	Apple Inc.	Portfolio asset

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AMZN	Amazon.com Inc.	Portfolio asset
JPM	JPMorgan Chase & Co.	Portfolio asset
MSFT	Microsoft Corporation	Portfolio asset
XOM	Exxon Mobil Corporation	Portfolio asset
SPY	SPDR S&P; 500 ETF	Benchmark

Daily returns were calculated from consecutive adjusted closing prices. The five individual stocks were used to construct the portfolios, while SPY was used as the benchmark for comparison.

4. Methodology

4.1 Daily Returns

A daily return measures the percentage change in an asset's price from one trading day to the next. It was calculated as:

Daily Return = (Today's Price / Previous Price) – 1

4.2 Annualised Volatility

Volatility measures how much returns fluctuate. Daily standard deviation was multiplied by the square root of 252, representing the approximate number of trading days in a year.

4.3 Sharpe Ratio

The Sharpe ratio measures return relative to volatility. A higher value indicates more historical return per unit of risk. A zero risk-free rate was assumed in this project.

4.4 Maximum Drawdown

Maximum drawdown measures the largest percentage decline from a previous portfolio peak to a subsequent trough. It provides a measure of historical downside risk.

4.5 Compound Annual Growth Rate

CAGR represents the constant annual growth rate that would produce the same overall investment growth over the period.

4.6 Minimum-Variance Optimisation

The minimum-variance portfolio seeks the combination of asset weights that produces the lowest estimated portfolio variance, subject to long-only weights and a total allocation of 100%. In simple terms, the optimisation attempts to find a combination of assets with the lowest historical fluctuation.

5. Historical Risk and Return Analysis

The historical analysis showed that the selected stocks had different levels of volatility. Amazon recorded the highest annualised volatility, while JPMorgan recorded the lowest among the five individual stocks.

Asset	Annualised Volatility
AAPL	28.14%
AMZN	36.73%
JPM	24.43%
MSFT	28.40%
XOM	26.69%
SPY	17.30%

The correlation analysis showed that Exxon Mobil had relatively low correlations with several technology stocks. This illustrates how assets from different sectors can behave differently and potentially contribute to diversification.

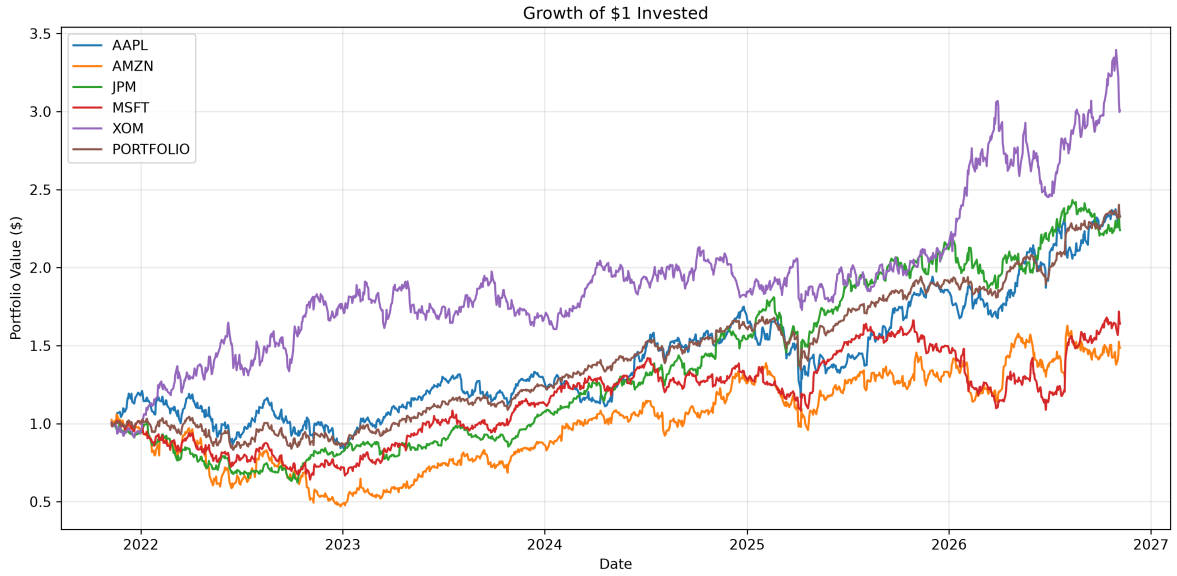


Figure 1. Historical growth of an initial investment of $1.

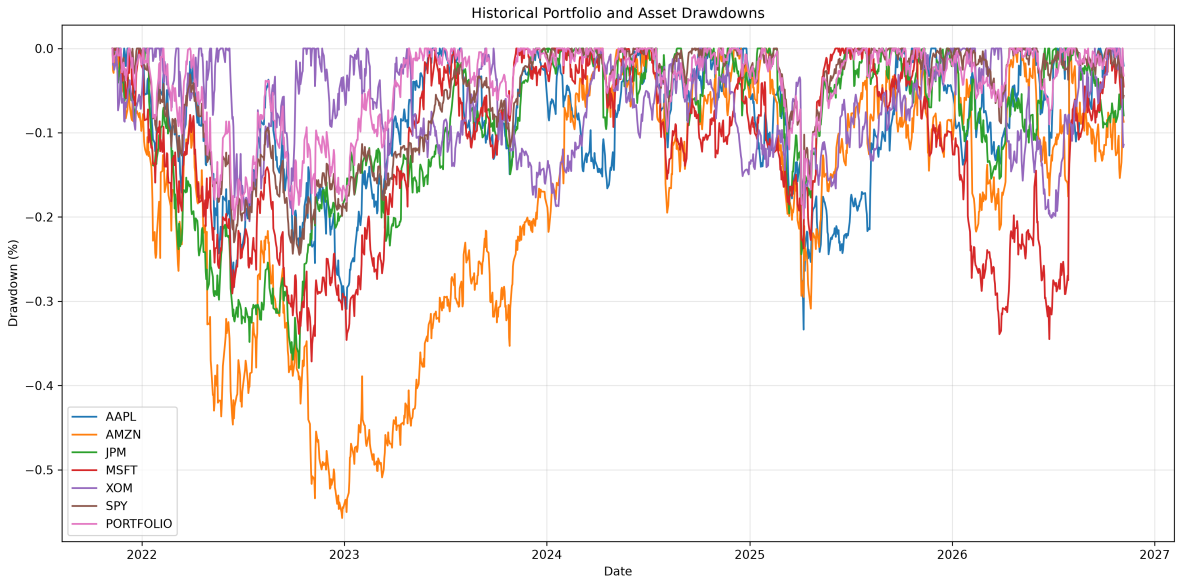


Figure 2. Historical drawdowns across the selected assets and portfolio.

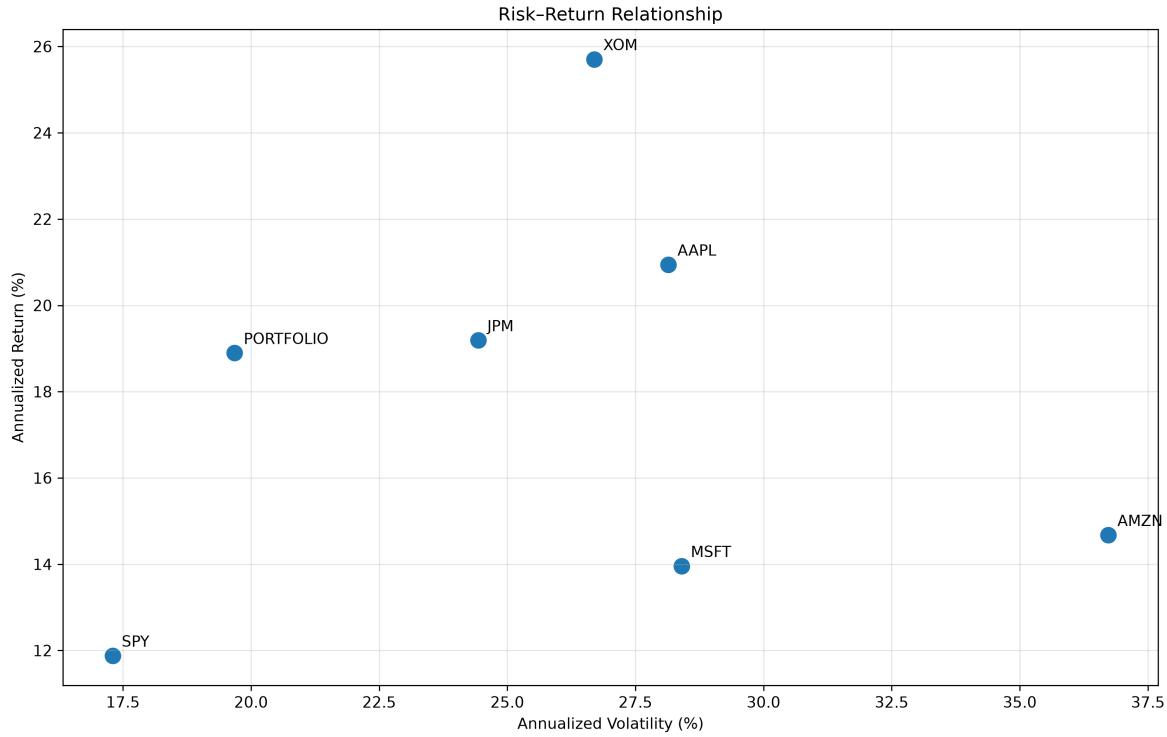


Figure 3. Relationship between historical annualised risk and return.

6. Portfolio Construction

6.1 Equal-Weight Portfolio

The equal-weight portfolio allocated 20% of its capital to each of the five stocks. It provides a straightforward benchmark for evaluating whether optimisation adds value.

6.2 Fixed Minimum-Variance Portfolio

The fixed minimum-variance portfolio used the training-period data from November 2021 through December 2024 to estimate the portfolio weights. Those target weights were then used during the out-of-sample test.

Asset	Training Weight
AAPL	14.54%
AMZN	0.00%
JPM	30.04%
MSFT	23.39%
XOM	32.02%

The zero allocation to Amazon means that the optimisation did not select it under the chosen minimum-variance objective and constraints. It does not mean that Amazon is inherently a poor investment.

6.3 Rolling Minimum-Variance Portfolio

The rolling strategy recalculated portfolio weights monthly using the previous 252 trading days of available data. This allowed the portfolio allocation to respond to changing historical risk and correlation estimates.

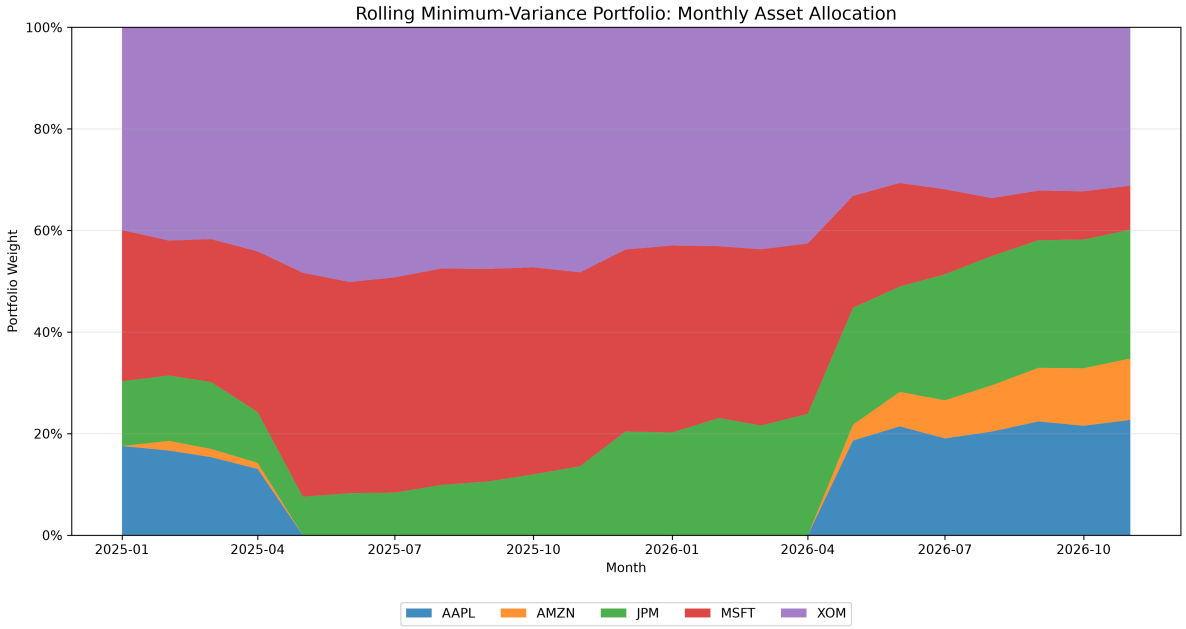


Figure 4. Changes in the rolling minimum-variance portfolio allocation.

7. Out-of-Sample Evaluation

The dataset was divided into a training period and a test period. The training period ended on 31 December 2024. The out-of-sample test began on 2 January 2025 and ended on 6 November 2026.

Out-of-sample testing is important because it evaluates a strategy on observations that were not used to estimate its original fixed portfolio weights. This provides a more realistic historical assessment than evaluating a strategy only on the data used to construct it.

7.1 Monthly Rebalancing

The strategies reset their target allocations at the beginning of each month. Weights were allowed to drift between rebalancing dates as asset prices changed. The SPY benchmark was held without portfolio rebalancing.

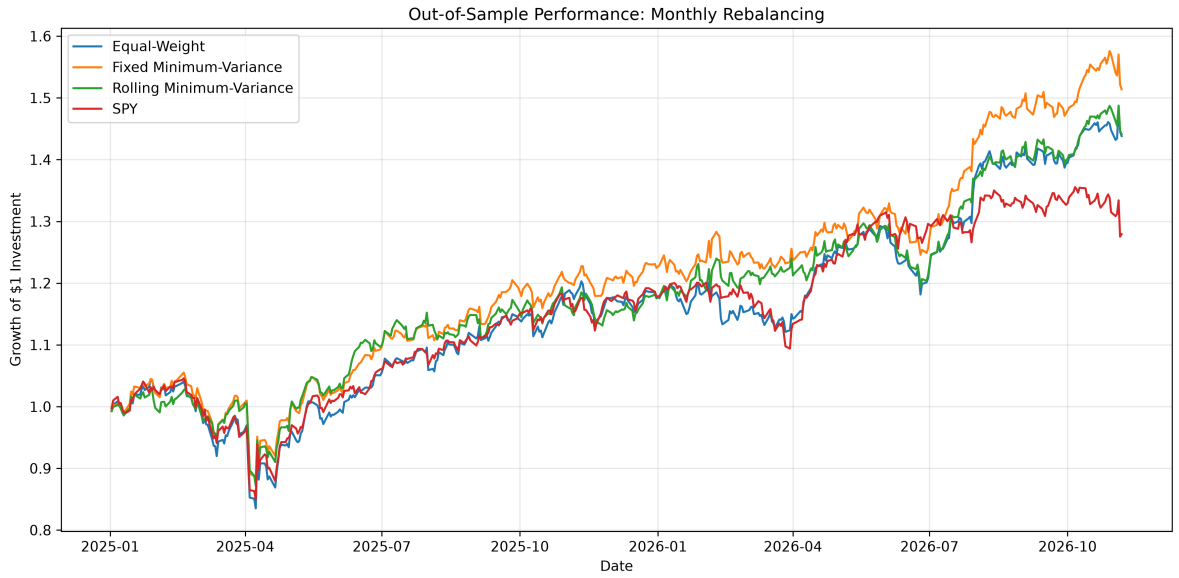


Figure 5. Out-of-sample growth of $1 under monthly rebalancing.

8. Out-of-Sample Performance Results

Portfolio	Annual Return	Volatility	Sharpe	Max DD	CAGR
Equal-Weight	21.38%	18.18%	1.18	–20.04%	21.82%
Fixed Min-Variance	23.91%	16.65%	1.44	–16.95%	25.26%
Rolling Min-Variance	21.27%	16.82%	1.26	–15.22%	21.96%
SPY	14.83%	17.10%	0.87	–18.76%	14.31%

The fixed minimum-variance strategy produced the highest CAGR and Sharpe ratio in this test. It also had the lowest annualised volatility. The rolling strategy recorded the smallest maximum drawdown, indicating the shallowest peak-to-trough decline among the strategies.

The equal-weight portfolio also performed competitively. Its results demonstrate that a simple allocation rule can remain a useful comparison against an optimisation-based approach.

9. Transaction Cost Analysis

Transaction costs were modelled at 0, 5, 10 and 25 basis points. One basis point is equal to 0.01%. The costs were applied to monthly portfolio turnover.

Cost	Equal-Weight CAGR	Fixed Min-Var CAGR	Rolling Min-Var CAGR
0 bps	21.82%	25.26%	21.96%
5 bps	21.80%	25.24%	21.92%
10 bps	21.78%	25.22%	21.88%

Cost	Equal-Weight CAGR	Fixed Min-Var CAGR	Rolling Min-Var CAGR
25 bps	21.73%	25.17%	21.77%

The rolling strategy had higher average monthly one-way turnover than the other two portfolios. Its average monthly turnover was approximately 5.01%, compared with 2.31% for equal-weight and 2.33% for fixed minimum-variance.

The tested transaction costs did not materially change the ranking of the strategies. However, the cost model is simplified and does not include all real-world trading expenses.

10. Estimation-Window Sensitivity

The rolling minimum-variance strategy was also evaluated using different historical estimation windows: 126, 252, 504 and 756 trading days. This examined whether the length of historical data used to estimate portfolio weights affected performance.

Window	Annual Return	Volatility	Sharpe	Max DD	CAGR
6 Months	17.60%	16.62%	1.06	–16.23%	17.61%
1 Year	21.27%	16.82%	1.26	–15.22%	21.96%
2 Years	22.34%	17.41%	1.28	–16.80%	23.15%
3 Years	23.22%	17.00%	1.37	–16.84%	24.33%

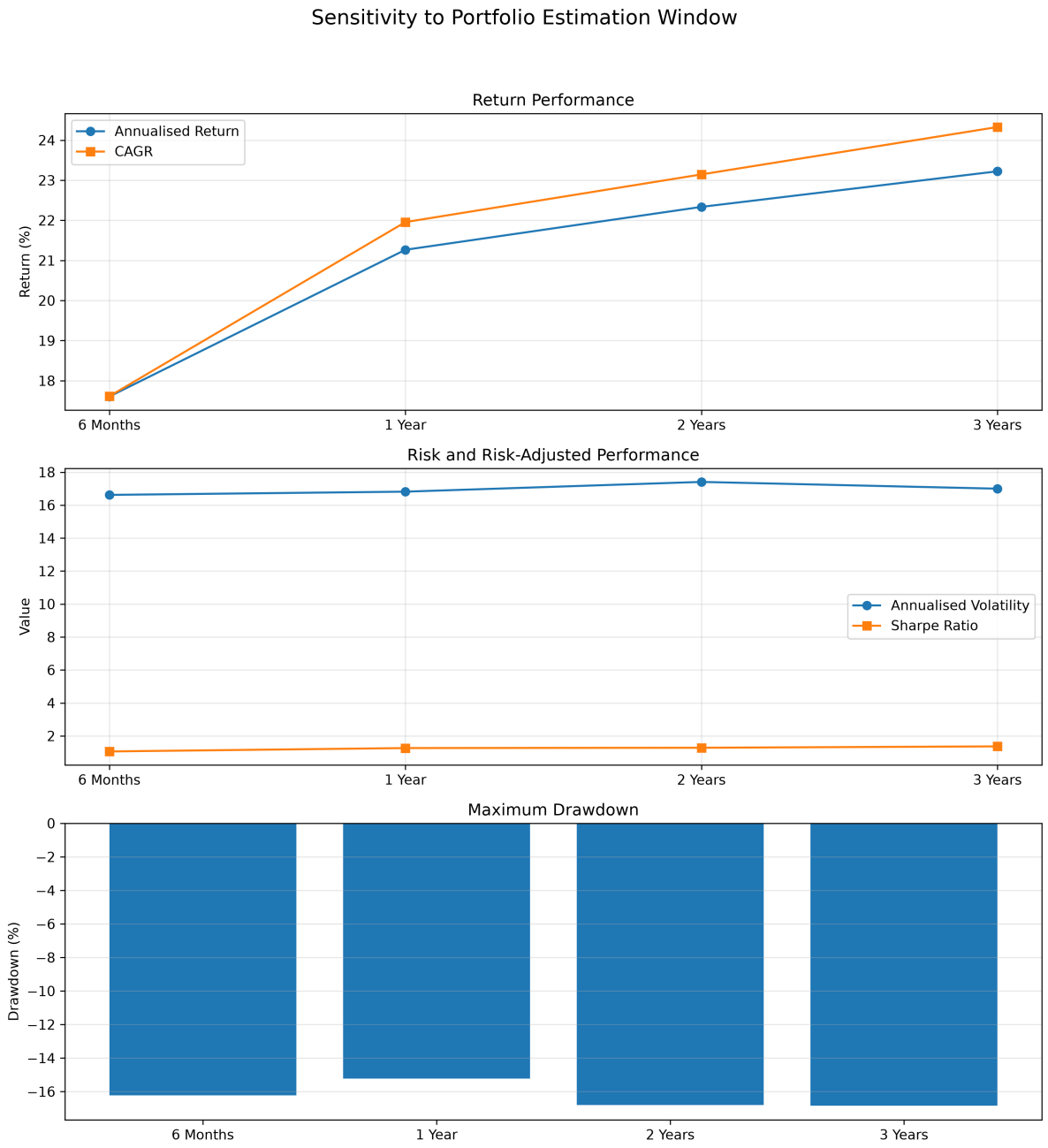


Figure 6. Performance sensitivity to the length of the estimation window.

The three-year estimation window recorded the highest CAGR and Sharpe ratio in this comparison. The one-year window produced the smallest maximum drawdown. These results are specific to the selected assets and test period.

11. Market-Condition Analysis

Out-of-sample trading days were classified according to the daily return of SPY. Days with a non-negative SPY return were classified as positive-market days, while days with a negative SPY return were classified as negative-market days.

This is a simple daily classification. It is not a formal definition of a bull market or bear market.

Market Condition	Equal-Weight	Fixed Min-Var	Rolling Min-Var	SPY
Positive Market	0.5833%	0.4468%	0.3774%	0.6963%
Negative Market	–0.5128%	–0.3270%	–0.2669%	–0.7055%

The minimum-variance portfolios experienced smaller average daily losses on negative SPY days than the benchmark. The rolling strategy had the smallest average loss on negative days. On positive days, however, the portfolios also recorded lower average daily returns than SPY.

12. Limitations

• The portfolio universe contains only five selected stocks.

• The test period represents one historical market environment.

• The Sharpe ratio assumes a zero risk-free rate.

• The transaction cost model excludes brokerage, bid-ask spreads, taxes and market impact.

• The analysis does not establish that the strategies will outperform in future periods.

• Historical covariance estimates may change substantially over time.

• The asset selection may introduce historical selection bias.

13. Conclusion

This project demonstrated how portfolio construction, risk measurement and out-of-sample evaluation can be combined in a structured financial analysis.

The fixed minimum-variance strategy delivered the strongest compound growth and risk-adjusted return in the selected out-of-sample period. The rolling minimum-variance strategy recorded the smallest maximum drawdown, while the equal-weight portfolio remained a competitive and relatively simple alternative.

The analysis also showed that portfolio performance depends on the estimation window, rebalancing frequency and trading cost assumptions. No single historical result should be treated as a guarantee of future performance.

Overall, the project provides a practical demonstration of financial data analysis using Python, including data preparation, return calculation, portfolio optimisation, risk measurement, backtesting and visualisation.

14. References and Tools

• Yahoo Finance – historical market price data: https://finance.yahoo.com/

• yfinance Python library – market data retrieval.

• pandas – data manipulation and analysis.

• NumPy – numerical computation.

• SciPy – numerical optimisation.

• Matplotlib – data visualisation.

Note: This report is an educational research project and does not constitute investment advice.